

M.A.S.H AI Agent Ecosystem

This document describes the 6 specialized AI agents running within the M.A.S.H (Medical Assistant & Services Hub) ecosystem. By splitting medical clinic workflows into modular, single-responsibility agents coordinated by the BandSDK, the platform removes clinical friction, automates administrative handoffs, and improves patient throughput.

1. Registration Agent

- **Role:** Smart Clinical Matchmaker & Directory Coordinator
 - **Responsibilities:**
 - Maintains a real-time index of active clinic clinicians, specialties, and schedules.
 - Analyzes the patient's symptoms/medical issues shared via the mobile chatbot.
 - Automatically matches and assigns the patient to the most suitable available specialist.
 - **Events:**
 - **Subscribes:** REQUEST_DOCTOR_MATCH
 - **Publishes:** DOCTOR_ASSIGNED, DOCTOR_DIRECTORY_UPDATED
 - **How it makes life easier:** Eliminates manual triage and scheduler bias. Patients are instantly assigned to the right care provider based on clinical need and schedule availability.
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2. Patient Management Agent

- **Role:** Queue & Remote Booking Orchestrator
- **Responsibilities:**
 - Processes remote appointment bookings made by patients from anywhere.
 - Manages live queues for patients based on scheduled times and real-time check-ins.
 - Handles cancellations and rescheduling slots dynamically.
- **Events:**

- **Subscribes:** BOOK_APPOINTMENT, PATIENT_CHECK_IN, APPOINTMENT_COMPLETED
 - **Publishes:** QUEUE_UPDATED, RESCHEDULE_CONFIRMED
 - **How it makes life easier:** Automates schedule updates from remote and in-clinic patient interactions, keeping the clinic queue balanced.
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3. Patient Navigation Agent

- **Role:** Wayfinding Assistant
 - **Responsibilities:**
 - Monitors doctor room allocations.
 - Calculates and sends routing paths and room guidance to the patient's mobile app upon check-in.
 - **Events:**
 - **Subscribes:** PATIENT_CHECKED_IN, DOCTOR_ROOM_CHANGE
 - **Publishes:** NAVIGATE_TO_ROOM
 - **How it makes life easier:** Guides patients directly to their doctor's door (e.g., "Room 101"), eliminating clinic navigation anxiety and lobby clutter.
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4. Patient Summary Agent

- **Role:** Clinical Record Aggregator
 - **Responsibilities:**
 - Queries historical patient records, lab tests, and past surgeries from Supabase.
 - Compiles an easy-to-read clinical summary on check-in.
 - **Events:**
 - **Subscribes:** GENERATE_SUMMARY (patient checked in event)
 - **Publishes:** SUMMARY_GENERATED
 - **How it makes life easier:** Saves clinicians hours of clicking through tabs in traditional EHR software by presenting a consolidated case brief upon check-in.
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5. Medicine/Prescription Management Agent

- **Role:** Prescription Safety & Order Auditor
- **Responsibilities:**

- Intercepts written clinical prescriptions.
 - Cross-references inventory stock levels with the Stock Management Agent.
 - Dispatches orders to pharmacy or handles out-of-stock prompts.
 - **Events:**
 - **Subscribes:** PROCESS_PRESCRIPTION
 - **Publishes:** ROUTE_TO_PHARMACY, REQUEST_HUMAN_INTERVENTION
 - **How it makes life easier:** Ensures doctors are alerted immediately if a drug is out of stock, prompting alternatives before the patient departs for the pharmacy.
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6. Stock Management Agent

- **Role:** Inventory & Logistics Controller
 - **Responsibilities:**
 - Monitors real-time stock levels of the clinic's medicine inventory.
 - Identifies patterns of high-frequency stock usage to suggest proactive reorders.
 - **Events:**
 - **Subscribes:** ROUTE_TO_PHARMACY (to deduct stock)
 - **Publishes:** TRIGGER_REORDER, STOCK_ALERTS_UPDATED
 - **How it makes life easier:** Automatically alerts pharmacists of low stock and flags repeating items for batch ordering, preventing critical drug shortages.
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Agent Handoff Interaction Flow

Below is a sequential flow showing how these agents collaborate asynchronously for a patient visit, starting from a remote symptom-based booking:

sequenceDiagram

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participant PatientApp as Patient Mobile
participant RegAgent as Registration Agent
participant PMAgent as Patient Management Agent
participant NavAgent as Patient Navigation Agent
participant SumAgent as Patient Summary Agent
participant RxAgent as Medicine Management Agent
participant StockAgent as Stock Management Agent
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Note over PatientApp: Remote Booking Flow (from anywhere)

PatientApp->>RegAgent: Book Appt: Sends symptoms (e.g., "Chest Pain")
activate RegAgent
RegAgent->>RegAgent: Match symptom to Specialty (Cardiologist) & check availability
RegAgent->>PMagent: Book slot with assigned specialist (e.g., Dr. Chen)
deactivate RegAgent
activate PMagent
PMagent-->>PatientApp: Appointment Confirmed (Dr. Chen, 09:00 AM)
deactivate PMagent

Note over PatientApp: In-Clinic Arrival Flow
PatientApp->>PMagent: Patient Checks In
activate PMagent
PMagent->>NavAgent: Broadcast NAVIGATE_TO_ROOM (Patient ID, Room 205)
deactivate PMagent

activate NavAgent
NavAgent-->>PatientApp: Display Guidance path to Room 205 (Dr. Chen)
deactivate NavAgent

PMagent->>SumAgent: Broadcast GENERATE_SUMMARY (Patient ID)
activate SumAgent
SumAgent->>SumAgent: Aggregate History, Vitals, Surgeries from Supabase
SumAgent-->>PMagent: Broadcast SUMMARY_GENERATED (Details loaded)
deactivate SumAgent

Note over PMagent: Doctor consults and writes prescription

PMagent->>RxAgent: Broadcast PROCESS_PRESCRIPTION (Rare-Antibiotic)
activate RxAgent
RxAgent->>StockAgent: Verify 'Rare-Antibiotic' in Stock
activate StockAgent
StockAgent-->>RxAgent: Out-of-Stock Status
deactivate StockAgent

RxAgent->>PMagent: Broadcast REQUEST_HUMAN_INTERVENTION (Alternative Required)
deactivate RxAgent
Note over PMagent: Doctor selects Ibuprofen as alternative

PMagent->>RxAgent: Broadcast PROCESS_PRESCRIPTION (Ibuprofen)
activate RxAgent
RxAgent->>StockAgent: Verify 'Ibuprofen' in Stock

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activate StockAgent
StockAgent-->>RxAgent: In-Stock Status
deactivate StockAgent
RxAgent->>StockAgent: Route Order & Deduct Stock
RxAgent-->>PatientApp: Prescription Sent to Pharmacy
deactivate RxAgent
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